



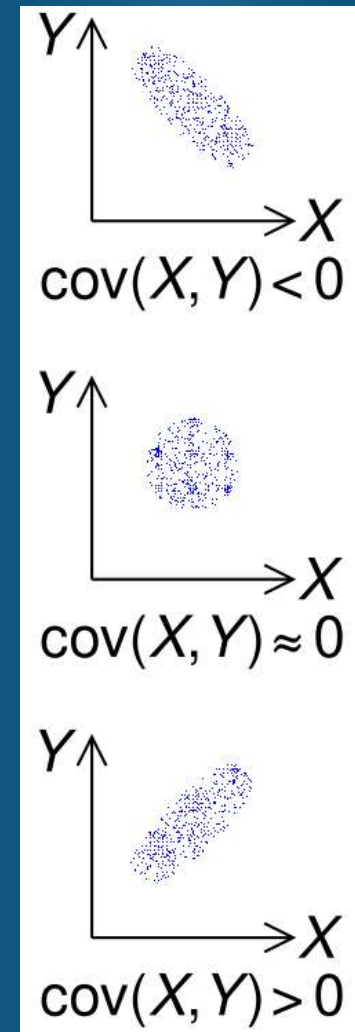
Covariance Matrix and Correlation Matrix

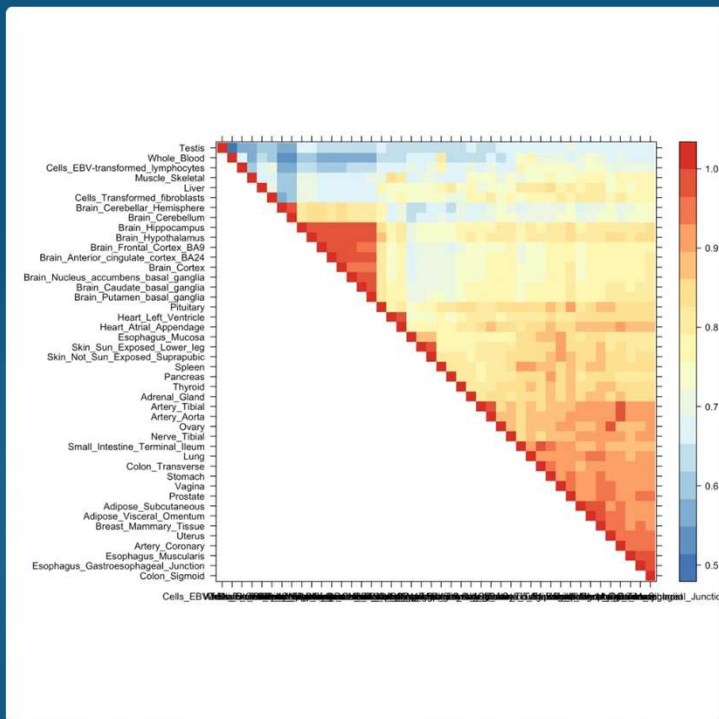
TS Nguyễn Quốc Minh

Covariance is a measure of the joint variability of two random variables.

For two jointly distributed real-valued random variables and with finite second moments, the covariance is defined as the expected value (or mean) of the product of their deviations from their individual expected values

$$\text{cov}(X, Y) = \text{E} [(X - \text{E}[X])(Y - \text{E}[Y])]$$





The diagonal elements of the matrix represent the **variances** of individual variables, while off-diagonal elements represent the covariances between pairs of variables.

The covariance matrix is always **symmetric**, as the covariance between variable A and B is the same as between B and A.

Calculating the Covariance Matrix

Step 1: Data Preparation

Step 2: Calculate Means

Step 3: Center the Data

Step 4: Compute Covariances

Step 5: Assemble the Matrix

- Mean of $X = \bar{X}$
- Mean of $Y = \bar{Y}$

- Covariance of X with X (Variance of X):

$$\text{Cov}(X, X) = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

- Covariance of Y with Y (Variance of Y):

$$\text{Cov}(Y, Y) = \frac{\sum (Y_i - \bar{Y})^2}{n - 1}$$

- Covariance of X with Y :

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Correlation (Pearson's correlation coefficient):

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where σ_X and σ_Y are the standard deviations of X and Y .

Covariance Matrix

$$\begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Cov}(Y, Y) \end{bmatrix}$$

Question:

When calculating the sample standard deviation (or sample variance), why do we divide by $(N-1)$ instead of N to get an **unbiased** estimate of the population standard deviation?

Bessel's correction

We are trying to estimate **POPULATION** (HUGE) variance based on a **SAMPLE** (SMALL) variance. So to account for that discrepancy, **$n-1$** is used, instead of **n**



Example: Calculating a Covariance Matrix

This dataset includes three variables:

- Years of Experience (X),
- Annual Performance Score (Y),
- Number of Projects Completed (Z)

Step 1: Collect the Data

Employee	Years of Experience (X)	Performance Score (Y)	Projects Completed (Z)
1	5	85	10
2	3	78	7
3	8	92	12
4	2	71	6
5	7	88	11

This dataset represents a sample of 5 employees.

Step 2: Calculate the Mean of Each Variable

1 Years of Experience (X)

$$\text{Mean}(X) = (5 + 3 + 8 + 2 + 7) / 5 = 5 \text{ years}$$

2 Performance Score (Y)

$$\text{Mean}(Y) = (85 + 78 + 92 + 71 + 88) / 5 = 82.8$$

3 Projects Completed (Z)

$$\text{Mean}(Z) = (10 + 7 + 12 + 6 + 11) / 5 = 9.2 \text{ projects}$$

Step 3: Compute the Deviations from the Mean

Employee	X - Mean(X)	Y - Mean(Y)	Z - Mean(Z)
1	0	2.2	0.8
2	-2	-4.8	-2.2
3	3	9.2	2.8
4	-3	-11.8	-3.2
5	2	5.2	1.8

Step 4: Calculate the Covariance Matrix

Compute Pairwise Products

Multiply the deviations for each pair of variables (X,Y), (X,Z), and (Y,Z) for each employee.

Sum Products

Add up the products for each variable pair across all employees.

Divide by (n-1)

Divide each sum by (n-1), where n is the number of data points (5 in this case).

Construct Matrix

Arrange the calculated covariances into a 3x3 symmetric matrix.

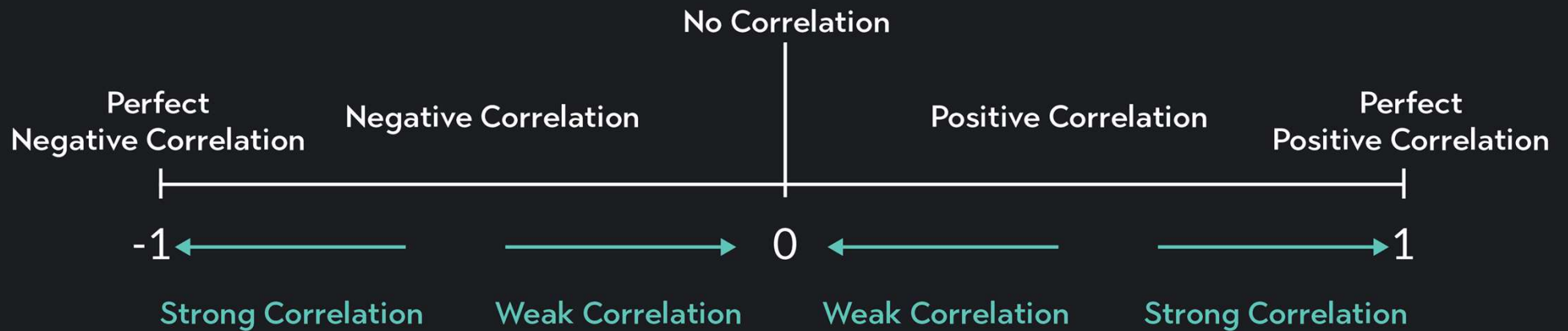
Correlation Matrix

- A correlation matrix contains Pearson correlation coefficients for all pairs of variables in a dataset. These coefficients range from -1 to 1, indicating the strength and direction of linear relationships.
- The correlation matrix is symmetric. Its diagonal elements are always 1, representing the perfect correlation of each variable with itself. Off-diagonal elements represent pairwise correlations.
- Correlation values close to 1 indicate strong positive relationships, values near -1 suggest strong negative relationships, and values around 0 imply weak or no linear relationship. This standardized scale makes interpretation more intuitive than covariance.
- The correlation matrix is widely used in exploratory data analysis, feature selection in machine learning, and in fields like psychology and social sciences where understanding relationships between variables is crucial. Its scale-invariant nature makes it particularly useful when dealing with variables measured in different units.

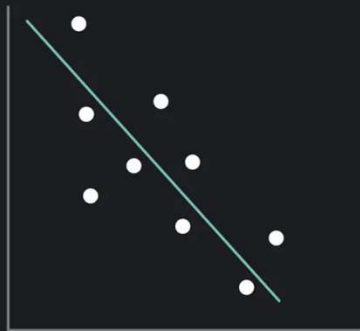
Correlation (Pearson's correlation coefficient):

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where σ_X and σ_Y are the standard deviations of X and Y .



Negative Correlation



The plotted points are clustered around a downward sloping line.

As the values one variable decreases, the other tend to increase and vice versa.

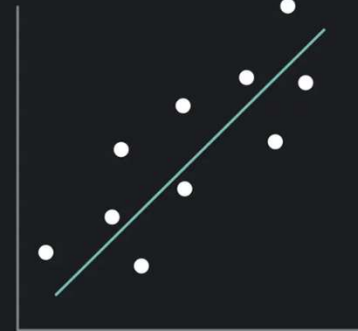
No Correlation



There is no linear trend.

There is no linear correlation between the two variables.

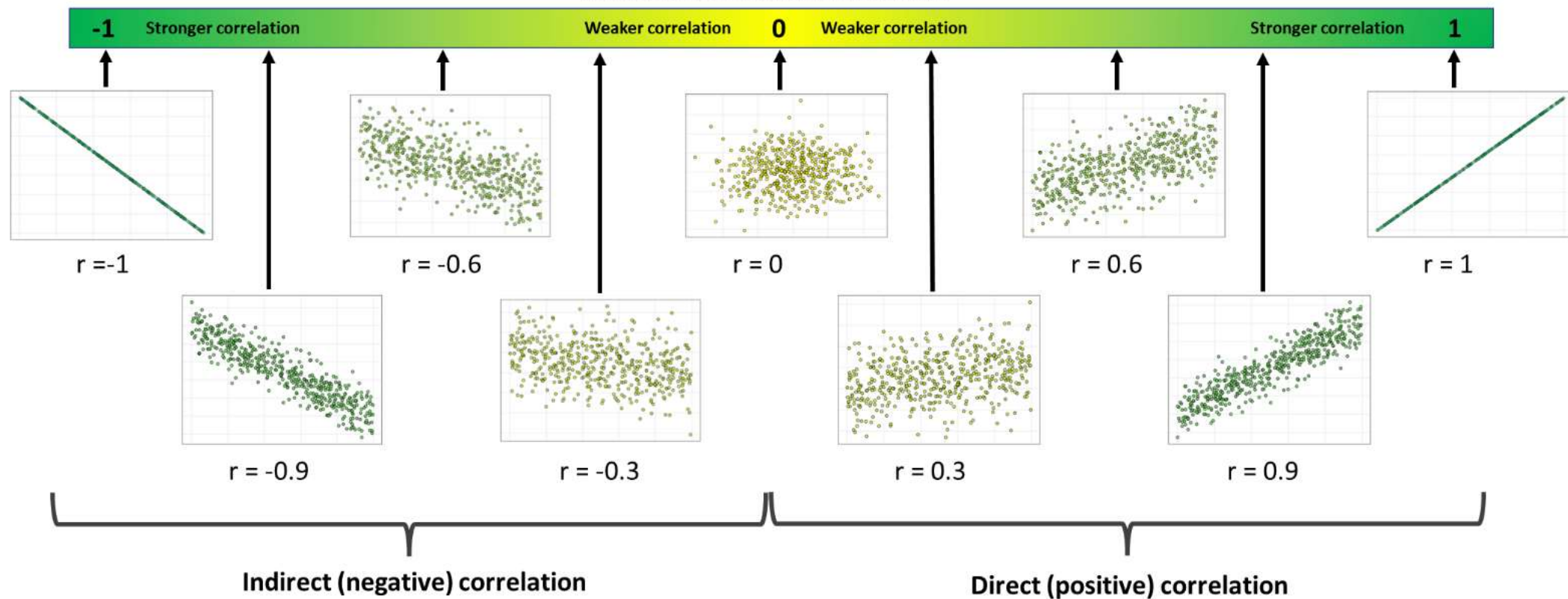
Positive Correlation



The plotted points are clustered around an upward sloping line.

As the values of one variable increase, the value of the other variable tends to increase as well.
As the value of one variable decreases, the value of the other also decreases.

Pearson correlation coefficient r



Interpreting the Correlation Matrix

Correlation Value	Strength of Relationship	Direction
0.00 to 0.19	Very weak	Positive
0.20 to 0.39	Weak	Positive
0.40 to 0.59	Moderate	Positive
0.60 to 0.79	Strong	Positive
0.80 to 1.00	Very strong	Positive

- **Covariance:** Measures the **joint variability** of two variables. If both variables tend to increase or decrease together, the covariance is positive. If one increases while the other decreases, the covariance is negative.
- **Correlation:** Measures the **strength and direction** of a linear relationship between two variables. Correlation standardizes covariance to a scale between -1 and 1, making it easier to interpret.

- **Covariance (for variables X and Y):**

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Here, $\bar{X}$ and $\bar{Y}$ are the means of X and Y .

- **Correlation (Pearson's correlation coefficient):**

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where σ_X and σ_Y are the standard deviations of X and Y .

- **Covariance:** Has **no fixed range** and depends on the scale of the variables. Large values of covariance indicate a strong relationship but are hard to interpret directly because they depend on the magnitude of the variables.
- **Correlation:** Always ranges from **-1 to 1**.
 - **+1:** Perfect positive linear relationship.
 - **-1:** Perfect negative linear relationship.
 - **0:** No linear relationship.

Covariance and Correlation

Covariance

Measures the degree to which two variables change together. It is scale-dependent and can range from negative infinity to positive infinity.

Correlation

A standardized measure of the linear relationship between two variables. It ranges from -1 to 1, with -1 indicating a perfect negative linear relationship, 0 indicating no linear relationship, and 1 indicating a perfect positive linear relationship.

Differences

Correlation is scale-independent and bounded between -1 and 1, making it easier to interpret and compare across different pairs of variables. Covariance, while providing similar directional information, is unbounded and scale-dependent, making it less suitable for direct comparisons between variable pairs with different units or scales.

Covariance:

The sign (positive or negative) indicates the direction of the relationship:

Positive covariance: Variables increase or decrease together.

Negative covariance: As one variable increases, the other decreases.

However, the magnitude of covariance does not provide standardized information about the strength of the relationship.

Correlation:

The sign indicates direction, while the magnitude indicates the strength:

A value close to $+1$ or -1 indicates a strong linear relationship.

A value close to 0 indicates a weak linear relationship.

Since correlation is standardized, it allows for easy comparison across datasets.

Feature	Covariance	Correlation
Definition	Joint variability of two variables	Strength and direction of linear relationship
Formula	$\frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$	$\frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$
Range	Unlimited	-1 to 1
Interpretation	Sign indicates direction	Sign indicates direction; magnitude indicates strength
Scale Dependence	Depends on scale	Independent of scale
Use Cases	Portfolio theory, variance analysis	Data analysis, feature selection in modeling

Calculating the Correlation Matrix

Step 1: Compute Covariance Matrix

Step 2: Calculate Standard Deviations

Step 3: Normalize Covariances

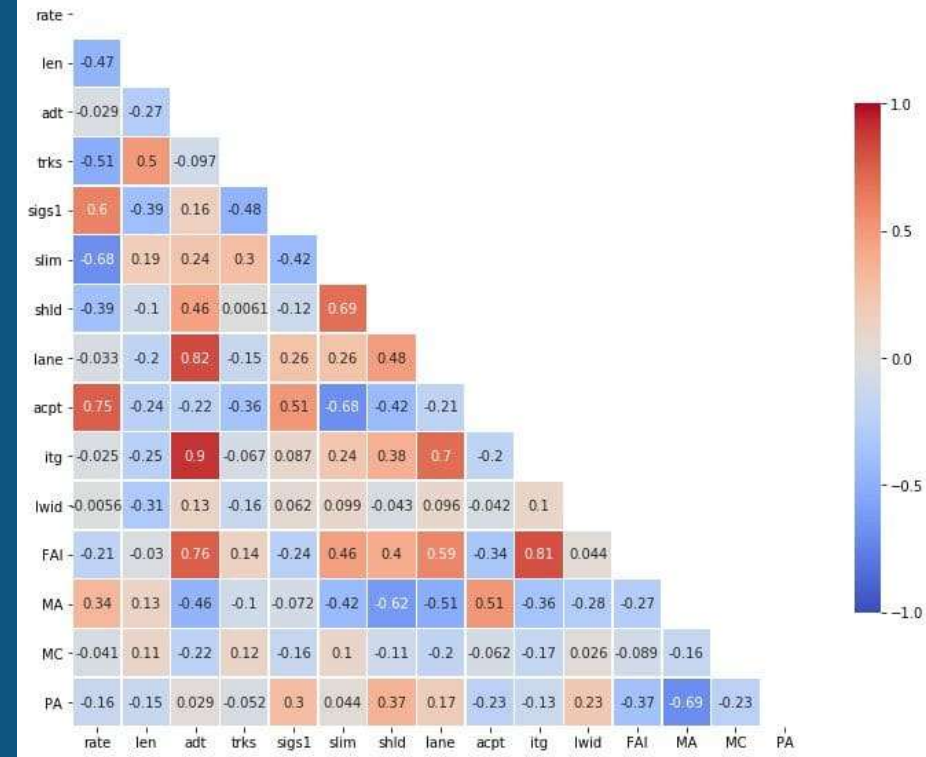
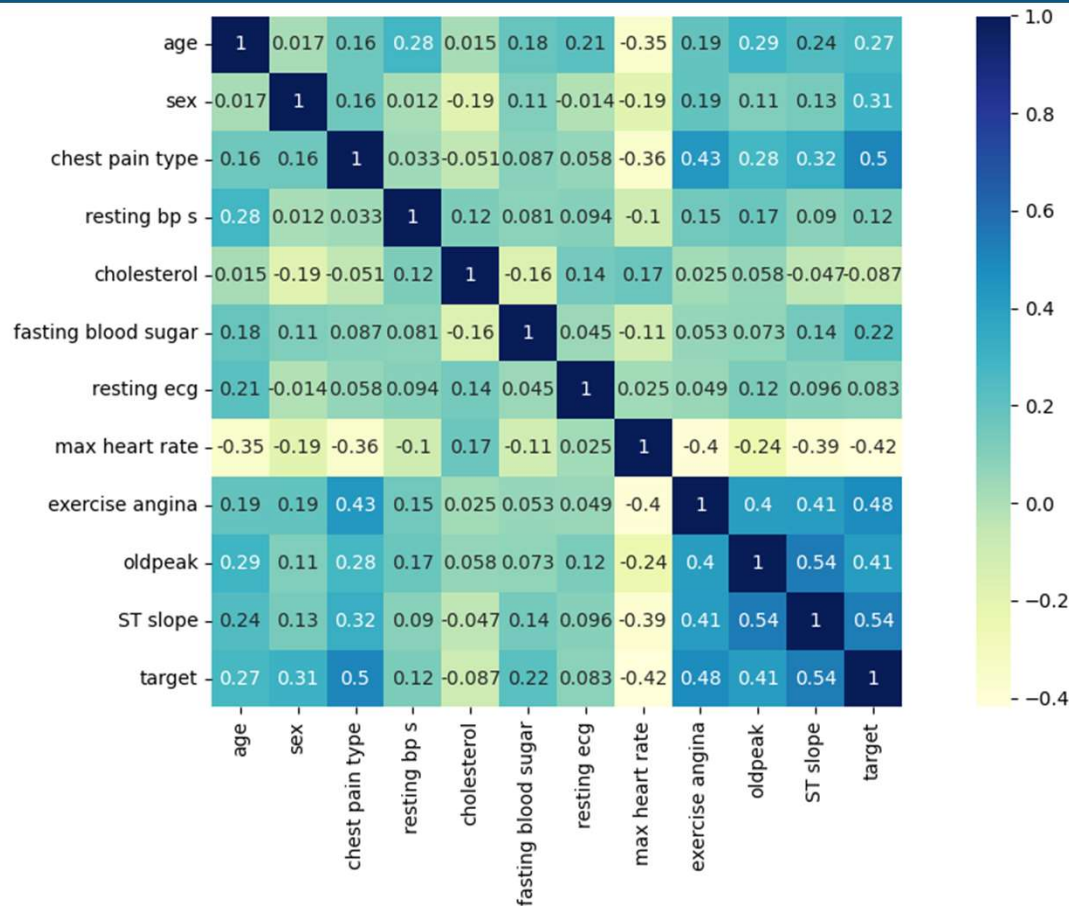
Step 4: Construct Correlation Matrix

Heatmap



- ▶ A heatmap correlation matrix is a 2D graphical representation that uses color to show the strength and direction of the relationships between variables in a dataset.
- ▶ Each cell in the matrix is colored based on its correlation coefficient, which ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no correlation.
- ▶ This visualization makes it easier to quickly identify patterns, such as strong positive or negative relationships between features.

Heatmap



Applications of correlation matrix/ heatmaps

- ▶ used in exploratory data analysis to visually identify relationships between variables in predictive modeling
- ▶ used in Feature selection:
 - identify which variables are strongly/highly correlated (positively or negatively)
 - group variables that move together
 - Simplify datasets by identifying redundant features for model optimization.